

ORIGINAL ARTICLE

Application of DNA barcoding techniques to mammal inventories in the African rain forest: droppings may inform us of the owners

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ABSTRACT The DNA barcoding of fecal samples was used to inventory large- and medium-sized terrestrial mammals in the African rain forest. First, the appropriate DNA region for the genetic identification of various species was determined. Although a region of mitochondrial cytochrome *c* oxidase I has been commonly used for DNA barcoding, the number of mammal species in which sequences of the region have been registered in the GenBank is limited, with the mitochondrial cytochrome *b* region (*cytb*) representing the most appropriate region for this type of analysis. Hence, genetic species identification was conducted using the mitochondrial *cytb* region for most samples, while the control region was used for small ungulates. We collected 259 fresh fecal samples of mammals in Moukalaba-Doudou National Park, Gabon. More than 70 % of analyzed samples produced sequences. The species was identified by examining the sequence identity between the sample and the identified species and that between the sample and the next closest species, or by constructing a phylogenetic tree containing closely related species. In total, the sequences of 19 mammal species were obtained. A smaller number of species were recorded by DNA barcoding compared to camera traps placed in the study area; however, DNA analyses were effective at discriminating morphologically similar species, such as small ungulates and carnivores, some of which were difficult to identify even when using camera traps. Genetic species identification using feces, in combination with direct observation and/or camera traps, can be useful for the accurate inventory of large- and medium-sized terrestrial mammals.

Key words: DNA barcoding, mammal inventory, noninvasive genetics, species identification

INTRODUCTION

Mammal inventory is important to investigate the biodiversity of certain regions, such as protected areas. However, it is difficult to record nocturnal and/or shy mammal species by direct observation. As a result, recent advances in mammal inventories have involved the use of camera traps (Trolle 2003, Yasuda 2004, Srbek-Araujo and Chiarello 2005, Gimán et al. 2007, Tobler et al. 2008, Nakashima 2015). Camera trapping has become one of the useful methods for monitoring large- and medium-sized terrestrial mammals. However, it is difficult to discriminate among species that have similar morphological characteristics, such as small carnivores and small ungulates. It is possible to use certain distinctive criteria to distinguish similar-looking species (e.g., Bahaa-El-Din et al. 2013); however, these criteria cannot be checked from some camera angles. Feces may provide another candidate for inventorying mammals, because feces are relatively easy to observe and collect, particularly for large- and medium-sized terrestrial mammals. Although it is easy to identify some mammal species

from the morphological aspects of feces, such as elephants and buffaloes, genetic species identification, including DNA barcoding, is necessary for other mammal species, such as small carnivores and small ungulates. Fortunately, recent improvements in noninvasive genetics have facilitated the analysis of DNA sequences using mammal feces (Taberlet et al. 1999, Nsubuga et al. 2004).

Genetic species identification using short DNA sequence has been conducted since the 1980's, with a significant increase the number of papers including the term “DNA barcode” since 2003 (Taylor and Harris 2012). This tool is powerful for taxa in which species identification by morphological traits alone is difficult because of the phenotypic plasticity of the trait, the existence of cryptic taxa, or phenotypic differences between sexes or among life stages (Valentini et al. 2009). Consequently, there is taxonomic bias in studies using DNA barcoding, with only a few studies focusing on mammals (Taylor and Harris 2012). For mammals, DNA barcoding has been mainly applied in rodents (Borisenko et al. 2008, Galan et al. 2012) and bats (Clare et al. 2007, Francis et al. 2010).

DNA barcoding may also represent a powerful tool for identifying species from a part of the animal (i.e., hair/fur, claws, or skin) or its droppings. For example, to monitor illegal trade, hair/fur samples in shaving brushes and in shahtoosh were used for the genetic identification of the Eurasian badger (*Meles meles*) and Tibetan antelope (*Pantholops hodgsonii*), respectively (Domingo-Roura et al. 2006, Lee et al. 2006). The method may also be applied to conservation biology. To monitor two sympatric endangered carnivores, the Amur leopard (*Panthera pardus orientalis*) and the Siberian tiger (*Panthera tigris altaica*), genetic species identification using fecal samples was conducted (Sugimoto et al. 2006). Genetic species identification using fecal samples was also conducted in tropical forests to investigate species distribution, their abundance, and their food (Miotto et al. 2007, Napolitano et al. 2008, van Vliet et al. 2008, Bowkett et al. 2009, Nakashima et al. 2010). These studies indicate the possibility of conducting mammal inventories using mammal feces.

The mitochondrial gene cytochrome *c* oxidase I (COI) has been frequently used for animal species after it was reported that the region could serve as a genetic barcode for animals (Hebert et al. 2003, Taylor and Harris 2012). Hebert et al. (2003) selected this region because the universal primers were very robust, and because COI appeared to contain a greater range of phylogenetic signal. However, in the forensic study of the Eurasian badger (*Meles meles*), species identification using the COI region failed, because COI reference sequence data did not exist for this species in the GenBank at the time of the study (Dawnay et al. 2007). When using genetic species identification based on feces, reference sequences must exist in the GenBank. However, the reference sequences of the COI region for some large- and medium-sized mammals may not exist, because of DNA barcoding focusing on just rodents and bats to date. Therefore, it is necessary to search the DNA region where sequences are already registered for most target mammals, before conducting genetic species identification using fecal samples.

In this study, we applied DNA barcoding to inventory mammals in an African rain forest in Gabon. First, the numbers of registered sequences of several mitochondrial regions for target mammal species was investigated, to select appropriate DNA regions. Second, we conducted genetic species identification using the fecal samples of species that were easily identified based on their morphological characteristic, to confirm the accuracy of this method. Third, genetic species identification using the fecal samples of species that were difficult to identify based on their morphology was conducted, such as small carnivores and small ungulates

which were abundant in the study area (Nakashima 2015). Finally, we discuss the utility of this method for mammal inventories and compared it with data collected by camera traps and direct observation.

MATERIALS AND METHODS

DNA database survey

We surveyed the number of registered sequences in target DNA regions for target species using the NCBI GenBank on September 27, 2013. The target species were the large- to medium-sized mammals that had been recorded in Gabon by Christy et al. (2008), excluding aquatic mammals and rodents (Appendix). Essentially, only species from which feces might be collected from the ground were selected. We included arboreal mammals because their feces might be collected. One of the target DNA regions was the COI, which has been used for many DNA barcoding studies. Other target DNA regions were three mitochondrial coding regions—cytochrome *b* (*cytb*) and two ribosomal RNA genes (16S rRNA and 12S rRNA)—because these regions have been used for the phylogenetic analyses of mammals (Allard and Honeycutt 1992, Johnson and O'Brien 1997, Hassanin and Douzery 1999, Castresana 2001, Quéroutil et al. 2001, Tougaard et al. 2001, Lecompte et al. 2002, Montgelard et al. 2002). The number of complete mitochondrial genome sequences or almost complete genome sequences of mitochondrion (the length was longer than 15,000 base pairs) that have been registered were also counted, because the sequences included all of the candidate genetic regions.

Sample collection and DNA analyses

We (mainly E.I.) collected the fecal samples of mammals in Moukalaba-Doudou National Park between September and November 2009 (totaling 42 study days). The national park is located in southern Gabon and consists of a mosaic of forest, savanna, and swamp. We followed one habituated gorilla group (Group Gentil; Ando et al. 2008, Inoue et al. 2013) on some days, and walked around a 30 km² area, mainly along with 6 phenology transects located in study area (Takenoshita et al. 2008), on other days, collecting fresh fecal samples using cotton swabs. We collected fecal samples not only in the forest but also in the savanna. Fecal samples were maintained in 2 ml lysis buffer solution containing 0.5 % SDS, 100 mM EDTA (pH 8.0), 100 mM

Tris-HCl (pH 8.0), and 10 mM NaCl (Hayaishi and Kawamoto 2006).

The feces of some mammals had species-specific morphological characteristics from which it was easy to identify the species. In Moukalaba, these animals included gorilla (*Gorilla gorilla*), chimpanzee (*Pan troglodytes*), elephant (*Loxodonta africana*), buffalo (*Syncerus caffer*), red river hog (*Potamochoerus porcus*), and waterbuck (*Kobus ellipsiprymnus*). For the mammal samples that could be easily identified, we analyzed the sequences of two samples from each species. In contrast, it was difficult to identify the species of carnivores, small ungulates, and rodents from fecal morphology. For these mammals, we analyzed the sequences of all the samples that we collected.

DNA extraction and sequencing analyses

DNA extraction was conducted using a QIAamp DNA Stool Mini Kit (Qiagen, Tokyo, Japan). For genetic species identification, primers L14841 and H15149 were used to amplify an approximate 300-bp fragment of the *cytb* gene (Table 1, Veron and Heard 2000). We used *cytb* in all cases, except for small ungulates. For carnivores, the sequencing results from some samples were contaminated, probably because the feces also contained the DNA of prey mammals. For samples where the sequencing results were unclear, *Panthera*-specific primers, PL1 and PH1 (Table 1, Nagata et al. 2005), were used, because leopards frequently eat other mammals. For small ungulates, the mitochondrial control region (around 675 bp) was used for species identification, because the region was reported to be superior in terms of identifying African duiker species compared to *cytb* (Table 1, Ntie et al. 2010).

Polymerase chain reaction (PCR) amplifications were carried out in a total 15- μ l volume, containing 2 μ l template, 150 μ M of each dNTP, 200 nM of each primer, 0.2 μ l of T4 gene 32 protein (Nippon Gene, Tokyo, Japan), and 0.5 U Expand High Fidelity enzyme mix (Roche Diagnostics,

Tokyo, Japan) in the PCR buffer supplied by the manufacturer. After initial incubation at 95°C for 5 min, 45 amplification cycles were performed with denaturation at 94°C for 30 s, annealing at 50–55°C (see Table 1) for 30 s, and extension at 72°C for 30 s. This process was followed by a final extension at 72°C for 30 min. PCR products were purified using ExoSap IT (USB, Cleveland, USA), and sequencing was carried out on an ABI3130xl (Applied Biosystems, Foster City, USA).

Genetic identification of species

The sequences were aligned and edited with MEGA5 (Tamura et al. 2011). Using NCBI BLAST embedded in MEGA5, the closest sequence to the target sequence was searched, and the percent identity of the two sequences was recorded. The identity of the target sequence to the next closest species sequence was also recorded. If there was a gap between the identity of the closest species sequence and that of the next closest species sequence, the closest species was identified as the target species. When the percent identity of the closest species sequence was low (less than 98%), a neighbor-joining phylogenetic tree, which contained closely related species sequences and one outgroup species sequence, was constructed using MEGA5 with Kimura 2-parameter model.

For central African duikers (Genera: *Philantomba* and *Cephalophus*), tree-based analyses that were constructed using control region sequences were used as the best method to identify species (Ntie et al. 2010). After editing the sequence using MEGA5, the closest sequence was searched by NCBI BLAST. We aligned all of the sequences that were close to the duiker sequences, and then constructed a neighbor-joining phylogenetic tree, which included the reference sequences of duikers, with sunis (*Neotragus moschatus*) as an outgroup (Ntie et al. 2010).

Table 1. Primers used for the genetic identification of species

Region	Primers	Sequence	Ta *	Reference
Cytochrome <i>b</i>	L14841	5'-CATCCAACATCTCAGCATGATGAAA-3'	50	Veron and Heard 2000
	H15149	5'-AAACTGCAGCCCCCTCAGAATGATATTGTCTCTCA-3'		
Cytochrome <i>b</i> for <i>Panthera</i>	PL1	5'-CYAAYAACCCTCAGGAATGGT-3'	55	Nagata et al. 2005
	PH1	5'-CGTTGTTTGGAWGTGTGGAGRGC-3'		
Control region	CR F1	5'-CTCCCTAAGGCTCAAGGAAGC-3'	55	Ntie et al. 2010
	CR R3	5'-CCTGAAGRAAGAACCAGATGTC-3'		

*Ta indicates annealing temperature

RESULTS

DNA database survey

We surveyed the registered sequences of 61 mammal species; namely, 20 Primates, 17 Carnivora, three Pholidota, one Tubulidentata, one Hyracoidea, one Proboscidea, and 18 Cetartiodactyla species (Appendix). Among these species, *cytb* regions were registered for 45 species (Table 2). Including mitochondrial genome information, information on *cytb* sequences was available for 53 species. The number of available species in the other regions was comparatively limited (Table 2). In particular, the number of registered carnivore species for the *cytb* region was larger compared to the other regions.

Genetic identification of species that are easily identified from fecal morphology

Twenty-five gorilla, 22 chimpanzee, 47 elephant, 31 buffalo, 13 red river hog, and 17 waterbuck fecal samples were collected. Two samples in each species were sequenced and clear sequences of the *cytb* region were successfully obtained from almost all the analyzed fecal samples. We failed to amplify the target region of just one sample (red river hog), hence a third sample of this species was sequenced. All of the analyzed sequences showed high similarity (more than 98 % identity) to the reference sequences of corresponding species and there was a gap between the sequence identity of the corresponding species and that of the next closest species (Table 3).

Genetic identification of rodents

One rodent fecal sample was collected, and a sequence in the *cytb* region was obtained from this sample. The percent identity of the closest species sequence (African brush-tailed porcupine, *Atherurus africanus*) was 98 %, and that to the next closest species (crested porcupine, *Hystrix cristata*) was low (85 %, Table 3). Although there was a lack of information on the closest species, the Asiatic brush-tailed Porcupine (*Atherurus macrourus*), in GenBank, the species from which our sample was derived was unambiguously identified as the African brush-tailed porcupine.

Genetic identification of carnivores

Thirty-one fecal samples were collected. Clear *cytb* sequences for 22 out of 31 samples were obtained using L14841 and H15149 primers. Nine samples failed, of which just one sample produced a clear sequence (Carnivore_24) using PL1 and PH1 primers. High identity (99–100 %) to the registered sequences confirmed that there was one sample of servaline genet (*Genetta servalina*), three samples (the same sequences) of long-snouted mongoose (*Herpestes naso*), four samples (the same sequences) of African civet (*Civettictis civetta*), two samples (two different sequences) of marsh mongoose (*Atilax paludinosus*), and eight samples (two sequences) of Egyptian mongoose (*Herpestes ichneumon*, Table 3). The sequence from Carnivore_24 was closest to the registered leopard sequence; however, the identity was not high (96 %, Table 3). A neighbor-joining phylogenetic tree was constructed using Carnivore_24 and several *Panthera* species sequences, with the domestic cat (*Felis catus*) sequence as the outgroup (Fig. 1). Carnivore_24 was

Table 2. Number of species for which reference sequences in each target region is available in DNA databank

Order	No. studied species	Cytb	COI	12s rRNA	16s rRNA
Primates	20	11 (16)	12 (14)	16 (17)	15 (17)
Carnivora	17	14	5 (6)	1 (2)	5
Pholidota	3	1 (2)	1 (2)	0 (1)	0 (1)
Tubulidentata	1	1	1	1	1
Hyracoidea	1	0 (1)	0 (1)	1	0 (1)
Proboscidea	1	1	0 (1)	0 (1)	0 (1)
Cetartiodactyla	18	17 (18)	14 (17)	16 (18)	9 (18)
Total	61	45 (53)	33 (42)	35 (41)	30 (44)

Complete or almost complete mitochondrial sequences of several species were registered. In such cases, we accessed all four target regions. The number including such instances is noted in parentheses.

Table 3. Identity of the sample sequences to the identified and next closest species by NCBI BLAST

Sample Name	Identified species		Next closest species	
	Species	Identity	Species	Identity
Buffalo_1	<i>Syncerus caffer</i>	98%	<i>Bubalus bubalis</i>	91%
Buffalo_2	<i>Syncerus caffer</i>	98%	<i>Bubalus bubalis</i>	92%
Chimp_1	<i>Pan troglodytes</i>	100%	<i>Pan paniscus</i>	96%
Chimp_2	<i>Pan troglodytes</i>	100%	<i>Pan paniscus</i>	97%
Elephant_1	<i>Loxodonta africana</i>	100%	<i>Mammuthus primigenius</i>	96%
Elephant_2	<i>Loxodonta africana</i>	100%	<i>Mammuthus primigenius</i>	97%
Gorilla_1	<i>Gorilla gorilla</i>	100%	<i>Homo sapiens</i>	90%
Gorilla_2	<i>Gorilla gorilla</i>	100%	<i>Homo sapiens</i>	90%
Hog_1	<i>Potamochoerus porcus</i>	99%	<i>Potamochoerus larvatus</i>	94%
Hog_2	<i>Potamochoerus porcus</i>	99%	<i>Potamochoerus larvatus</i>	94%
Waterbuck_1	<i>Kobus ellipsiprymnus</i>	99%	<i>Kobus kob</i>	91%
Waterbuck_2	<i>Kobus ellipsiprymnus</i>	99%	<i>Kobus kob</i>	91%
Carnivore_1	<i>Genetta servalina</i>	100%	<i>Genetta cristata</i>	99%
Carnivore_2	<i>Herpestes naso</i>	99%	<i>Herpestes urva</i>	93%
Carnivore_4	<i>Herpestes naso</i>	99%	<i>Herpestes urva</i>	93%
Carnivore_16	<i>Herpestes naso</i>	99%	<i>Herpestes urva</i>	93%
Carnivore_3	<i>Civettictis civetta</i>	100%	<i>Viverra megaspila</i>	90%
Carnivore_18	<i>Civettictis civetta</i>	100%	<i>Viverra megaspila</i>	90%
Carnivore_19	<i>Civettictis civetta</i>	100%	<i>Viverra megaspila</i>	90%
Carnivore_20	<i>Civettictis civetta</i>	100%	<i>Viverra megaspila</i>	90%
Carnivore_6	<i>Atilax paludinosus</i>	100%	<i>Herpestes ichneumon</i>	93%
Carnivore_14	<i>Atilax paludinosus</i>	99%	<i>Herpestes ichneumon</i>	93%
Carnivore_8	<i>Herpestes ichneumon</i>	99%	<i>Galerella sanguinea</i>	93%
Carnivore_13	<i>Herpestes ichneumon</i>	99%	<i>Galerella sanguinea</i>	93%
Carnivore_15	<i>Herpestes ichneumon</i>	99%	<i>Galerella pulverulenta</i>	92%
Carnivore_17	<i>Herpestes ichneumon</i>	100%	<i>Galerella pulverulenta</i>	93%
Carnivore_22	<i>Herpestes ichneumon</i>	100%	<i>Galerella pulverulenta</i>	93%
Carnivore_26	<i>Herpestes ichneumon</i>	99%	<i>Galerella sanguinea</i>	93%
Carnivore_28	<i>Herpestes ichneumon</i>	100%	<i>Galerella pulverulenta</i>	93%
Carnivore_29	<i>Herpestes ichneumon</i>	99%	<i>Galerella sanguinea</i>	93%
Carnivore_24	<i>Panthera pardus</i>	96%	<i>Panthera tigris</i>	92%
Carnivore_5*	<i>Caracal aurata</i>	96%	<i>Panthera tigris</i>	91%
Carnivore_7*	<i>Caracal aurata</i>	96%	<i>Panthera tigris</i>	91%
Carnivore_21*	<i>Caracal aurata</i>	96%	<i>Panthera tigris</i>	91%
Carnivore_27*	<i>Caracal aurata</i>	96%	<i>Panthera tigris</i>	91%
Rodent_1	<i>Atherurus africanus</i>	98%	<i>Hystrix cristata</i>	85%

*The identity of the closest sequences was low by NCBI BLAST analyses (91% to *Acinonyx jubatus*). The identities to the sequences of *Caracal* species (described in the supporting materials of Johnson et al. 2006) are shown in this table.

clustered together with the leopard sequences (*Panthera pardus orientalis*), hence we concluded that the fecal sample belonged to the African leopard (*Panthera pardus pardus*). The sequences of Carnivore_05, 07, 21, and 27 were identical, with the identity of the registered sequences being low (91 %, Table 3). From the fecal size, the feces

were inferred to be those of the African golden cat (*Caracal aurata*). No sequences of this species were found in the NCBI GenBank; however, the sequence information was available in the supporting materials of Johnson et al. (2006). A neighbor-joining phylogenetic tree was constructed using our sequences, while the sequences of the African

golden cat and close species were retrieved from Johnson et al. (2006). The samples were clustered together with the African golden cat (Fig. 2).

Genetic identification of small ungulates

We collected 72 fecal samples. The control region sequences were successfully obtained from 60 out of 72 fecal samples. Thirty-one different sequences were obtained from 60 samples. NCBI BLAST results showed that five fecal samples (three different sequences) belonged to water chev-

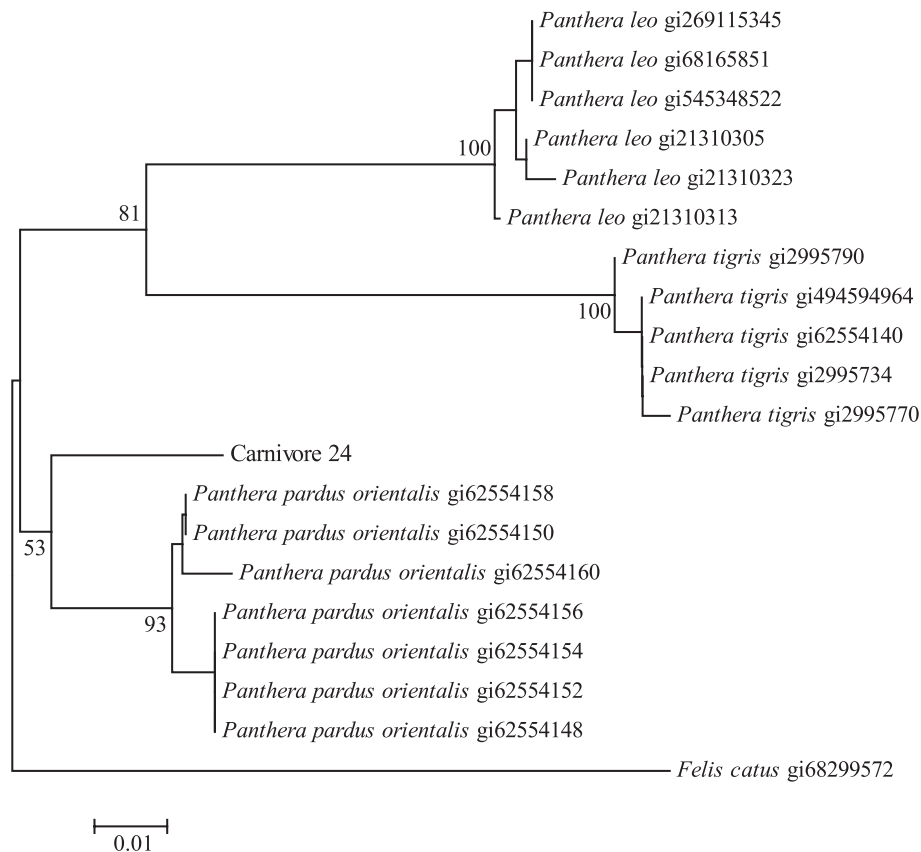


Fig. 1. Median-joining tree of mitochondrial cytochrome *b* region of *Panthera* species and our sample (Carnivore_24), using the domestic cat (*Felis catus*) as an outgroup. Bootstrap values based on 10,000 replicates are shown above the relevant branches.

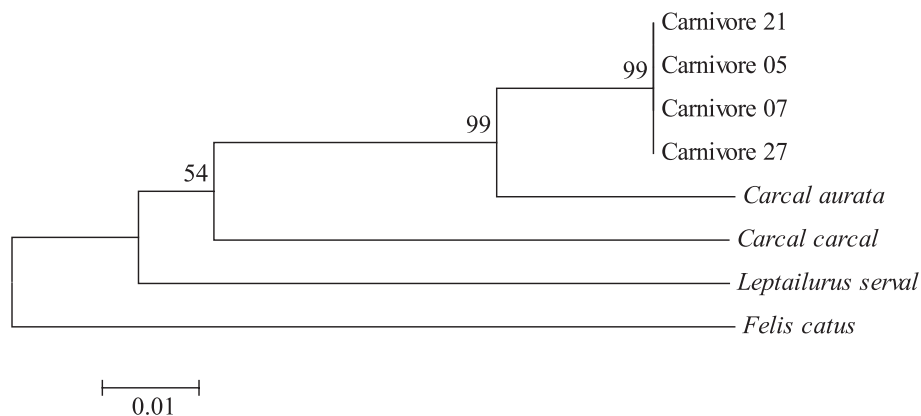


Fig. 2. Median-joining tree of mitochondrial cytochrome *b* region of the golden cat and closely related species, which were described in Johnson et al. (2006), and our samples (Carnivore_05, 07, 21, 27), using the domestic cat (*Felis catus*) as an outgroup. Bootstrap values based on 10,000 replicates are shown above the relevant branches.

rotain (*Hyemoschus aquaticus*) with high similarity (99–100% identity). We concluded that the sample belonged to water chevrotain because the identity of these three sequences to the next closest species did not exceed 90%. The closest species of the other 55 samples (28 different sequences) were duiker sequences (Genera: *Philantomba* and

Cephalophus). The species identification of the duiker samples was conducted using a phylogenetic tree (Fig. 3). The phylogeny showed that there were 40 samples (21 different sequences) of blue duikers (*Philantomba monticola*), five samples (two sequences) of yellow-backed duikers (*Cephalophus silvicultor*), one sample of bay duiker (*Cephalophus*

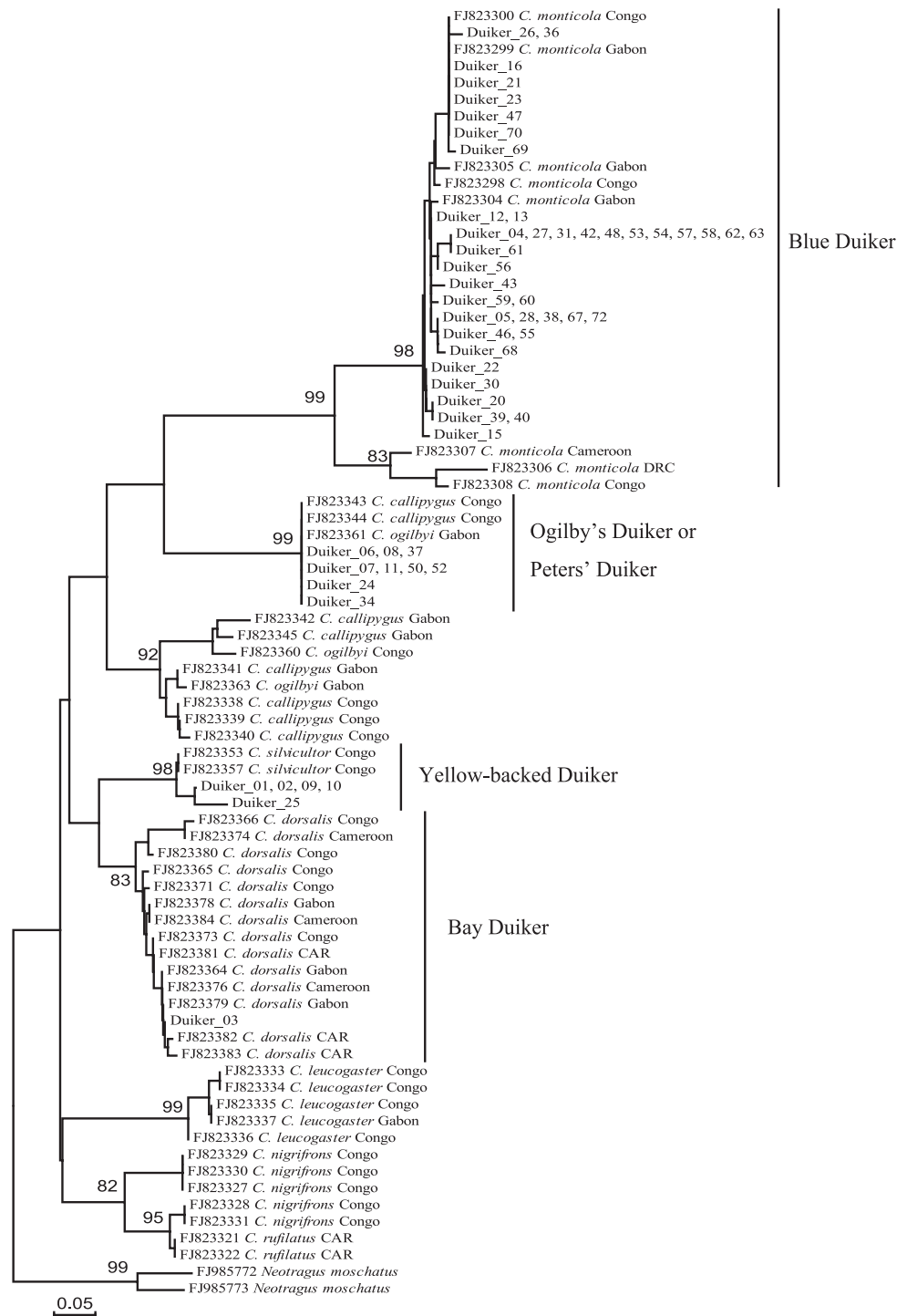


Fig. 3. Median-joining tree of the mitochondrial control region of duikers and our samples (Duiker_01–73), using the suni (*Neotragus moschatus*) as an outgroup. Bootstrap values based on 10,000 replicates are shown above the relevant branches.

dorsalis), and nine samples (four sequences) of Ogilby's (*Cephalophus ogilbyi*) or Peters' duikers (*Cephalophus calipygus*).

DISCUSSION

DNA region for species identification using unknown samples

The genomic information of target species in the DNA database is required to identify unknown species samples. A survey of the DNA database showed that a part of the *cytb* region was appropriate, at least at the time of study, for the identification of mammal species in Gabon, particularly carnivores (Table 2). However, information about some mammal species was not available, even with *cytb*. More genetic information must be collected to identify the species belonging to the unknown samples. In such cases, any mitochondrial gene may be a candidate region, because Luo et al. (2011) found no significant difference in intra- or inter-specific variability among mitochondrial genes using several mammals. One important issue when selecting target regions is the universal primers. COI may be a candidate region because universal primers may be applied to most, if not all, animal phyla (Herbert et al. 2003). This study showed that the *cytb* primers (L14841 and H15149) are useful for all target mammals in Gabon. Recent genetic improvements have facilitated the sequencing of the whole mitochondrial genome. Therefore, more genetic information will be available for identifying species using unknown samples in the future.

Success rate of sequencing using fecal samples

PCR amplification using fecal samples may be difficult because of low DNA concentration and the presence of PCR inhibitors (Waits and Paetkau 2005). The sequences of various mammals were obtained using a single method in this study. The success rate of sequence determination was quite high; specifically, 74 % (23/31) for carnivores, 83 % (60/72) for small ungulates, and 93 % (13/14) for all other mammals. These results indicate that almost fresh mammal feces provide DNA sequences using our method.

Accuracy of species identification

The genetic identification of species that are easy to

identify from fecal morphology confirmed the accuracy of the method. We examined the difference between the sequence identity of the sample to the identified species and that to the next closest species (Table 2). A difference was found in most examined species, with high identity to the identified species. In chimpanzees and elephants, the identity of the next closest species was high; however, there was a gap between the identity to the identified species and that to the next closest species. It was not possible to mis-identify these species because the distribution of the second candidate species of chimpanzees was different (bonobos, *Pan paniscus*), while the second candidate species of elephants is extinct (mammoth, *Mammuthus primigenius*).

For leopards and golden cats, the sequence identity to the identified species was not high (96 %). The sequence data of close species was available, and the phylogenetic tree using these data confirmed the same cluster with the corresponding species. In addition, the camera trap analyses (Table 4, Nakashima 2015) also confirmed the existence of these species. Low identity to the identified species was caused by their geographical distribution and the availability of limited genetic data. In the case of the leopard, the registered sequences, including subspecies information, were those of only Amur leopards (*Panthera pardus orientalis*). Their sequences might be slightly different to the African leopard subspecies (*Panthera pardus pardus*). For golden cats, the wide geographical distribution and limited genetic information might influence the low identity. The geographical sampling site from which the reference sequence was obtained was probably far from our sampling site, which caused intra-genetic variation. In addition, in African buffaloes and the African brush-tailed porcupines, which are widely distributed across equatorial Africa, and which have limited registered sequences, the identity to the identified species was slightly low (98 %, Table 2). This result indicates that the identity to the identified species depends on the volume of registered sequence data.

The two duiker species, Ogilby's and Peters' duikers, were not identified, even when using a phylogenetic tree (Fig. 3). There was low genetic variation (less than 1 %) among all four sequences that were identified as Ogilby's or Peters' duikers in this study. Previous studies indicate that these two species have high genetic similarity in the mitochondrial genome, along with it being difficult to identify these species using mitochondrial genetic information (Ntie et al. 2010, Hassanin et al. 2012). Hassanin et al. (2012) suggested that these two species, including another similar species (*Cephalophus weynsi*), might be subspecies, rather than species rank status. Nucleic genetic information and the reassessment of the taxonomy of these duikers are required.

Table 4. Mammal inventory by DNA barcoding compared to that obtained using camera traps and direct observation in Moukalaba-Doudou National Park

Order	Scientific name	Common name	Camera ¹	Direct ²	DNA
Primates	<i>Pan troglodytes</i>	Chimpanzee	+	+	+
	<i>Gorilla gorilla</i>	Western gorilla	+	+	+
	<i>Mandrillus sphinx</i>	Mandrill	+	+	
	<i>Cercocebus torquatus</i>	Red-capped mangabey	+	+	
	<i>Lophocebus albigena</i>	Grey-cheeked mangabey		+	
	<i>Miopithecus ogouensis</i>	Gabon talapoin		+	
	<i>Cercopithecus pogonias</i>	Crowned monkey		+	
	<i>Cercopithecus nictitans</i>	Putty-nosed monkey		+	
	<i>Cercopithecus cephus</i>	Moustached monkey		+	
	<i>Euoticus elegantulus</i>	Western needle-clawed galago	+	+	
	<i>Galago</i> spp. ³		+		
Carnivora	<i>Hydrichtis maculicollis</i>	Spotted-necked otter		+	
	<i>Bdeogale nigripes</i>	Black-legged mongoose	+		
	<i>Herpestes ichneumon</i>	Egyptian mongoose	+	+	+
	<i>Herpestes naso</i> ⁴	Long-snouted mongoose	+	+	+
	<i>Atilax paludinosus</i> ⁴	Marsh mongoose	+		+
	<i>Civettictis civetta</i>	African civet	+		+
	<i>Genetta servalina</i>	Servaline genet	+		+
	<i>Genetta maculata</i>	Rusty-spotted Genet	+		
	<i>Poiana richardsonii</i>	African linsang	+		
	<i>Nandinia binotata</i>	African palm civet	+		
	<i>Profelis aurata</i>	African golden cat	+		+
	<i>Panthera pardus</i>	Leopard	+	+	+
Pholidota	<i>Smutsia gigantea</i>	Giant pangolin	+	+	
	<i>Uromansis tetradactyla</i> ⁵	Long-tailed pangolin	+		
	<i>Phataginus tricuspis</i> ⁵	Tree pangolin	+		
Proboscidea	<i>Loxodonta africana</i>	African elephant	+	+	+
Cetartiodactyla	<i>Potamochoerus porcus</i>	Red river hog	+	+	+
	<i>Hyemoschus aquaticus</i>	Water chevrotain	+	+	+
	<i>Syncerus caffer</i>	African buffalo	+	+	+
	<i>Tragelaphus spekii</i>	Sitatunga		+	
	<i>Tragelaphus scriptus</i>	Bushbuck	+	+	
	<i>Philantomba monticola</i>	Blue duiker	+	+	+
	<i>Cephalophus silvicultor</i>	Yellow-backed duiker	+	+	+
	<i>Cephalophus dorsalis</i> ⁶	Bay duiker	+	+	+
	<i>Cephalophus dorsalis</i> ⁶	Peters' duiker	+	+	+
	<i>Cephalophus ogilbyi</i> ⁶	Ogilby's duiker			
	<i>Kobus ellipsiprymnus</i>	Waterbuck	+	+	+

^{1,2} Data of camera traps and direct observation was modified from the data in Nakashima (2015).

³⁻⁶ Species with the same superscript number could not always be distinguished from one another even when using camera traps.

The most problematic case was the servaline genet sample (carnivore_01, Table 3). The sequence was identical to the sequence of servaline genets (*Genetta servalina*). However, the identity to the next closest species, crested genets (*Genetta cristata*), was 99 %. Gaubert and Begg (2007) showed that the two registered sequences of crested

genets were actually chimeric sequences, and suggested that this phenomenon might be caused by the experimental procedure of extracting skin samples from museum specimens. Therefore, the species in our sample was identified as a servaline genet. Hybridization between these two species has also reported (Gaubert et al. 2006); therefore, it was

difficult to conclude the species based on the genetic data from just one sample.

The high accuracy in species identification in this study was because the sequence data of almost small carnivores and ungulates was available, because detailed phylogenetic studies of genets (Gaubert et al. 2004) and mongooses (Veron et al. 2004) have been conducted, and because of comprehensive species diagnostic studies of small ungulates (Ntie et al. 2010). However, sufficient genetic information was not available for widely distributed species, such as leopards, golden cats, buffaloes, and porcupines. In the GenBank, the information on the original geographic location of many sequence samples was missing. More sequence data, including sample geographic location, are required for accurate species identification and conservation practice, such as illegal trade detection using genetic techniques (Dawnay et al. 2007).

Comparison with direct observation and camera traps

The sequences of 19 mammal species were obtained in this study. It is often difficult to record nocturnal and/or shy mammal species by direct observation. Compared to direct observation data in the same study site, Moukalaba (Nakashima 2015), several species of small carnivores were only identified by DNA barcoding (Table 4). In addition, DNA barcoding identified the species which would have been difficult to identify accurately by direct observation alone, such as mongoose, genet, and duiker species. Therefore, DNA barcoding, which has the potential to provide the accurate species information, represents a useful tool to inventory mammals, especially small carnivores. On the other hand, arboreal primates such as mangabeys and guenons were recorded by direct observation, but we did not collect their fecal samples probably due to their arboreal ecology. The disadvantage for arboreal species might be important to be considered when applying DNA barcoding to mammal inventory.

A smaller number of species were recorded by DNA barcoding compared to camera traps in this study (Table 4, Nakashima 2015). Most of the species that were only recorded by camera traps were arboreal carnivores and pangolins. One of the reasons why only a limited number of species were identified by DNA barcoding was the effort required for sample collection. Only 42 study days by primarily one researcher were required to collect the feces from 19 mammal species. The feces in open area are easy to find compared with those in dense forest and feces containing fibers remain longer period. Therefore, the species

which defecated in open area or deposited feces with fibers might be easily collected and only a few efforts might be needed to collect them. Greater effort to search feces, such as searching the dense forest carefully, is required in order to increase the number of species identified by DNA barcode.

In addition, some species were difficult to discriminate even when using camera traps, but were identified by DNA barcode. For instance, two mongooses (long-snouted mongoose, *Herpestes naso*; marsh mongoose, *Atilax paludinosus*) and three red duikers (bay duiker, *Cephalophus dorsalis*; Peters' duiker, *Cephalophus callipygus*; Ogilby's duiker *Cephalophus ogilbyi*) cannot be always distinguished by camera traps. However, it was also not possible to discriminate the two duiker species (Peters' and Ogilby's duiker) by DNA analyses, because of their similarity in mitochondrial genome (Ntie et al. 2010, Hassanin et al. 2012). Ultimately, species identification by DNA barcode probably has the advantage of discriminating even morphologically similar species, as long as sufficient data are available. In conclusion, DNA barcoding of fecal samples, in combination with direct observation and/or camera traps, may be a powerful tool for accurate mammal inventory. This method may be useful in the place where there has not been intensive research and there exists several species of small carnivores and ungulates.

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Appendix 1. Number of mitochondrial gene sequences in the GenBank for terrestrial mammals recorded in Gabon

Order	Scientific name	Common name	IUCN Status	Cytb	COI	12s rRNA	16s rRNA	Mt genome	Partial Mt
Primates	<i>Pan troglodytes</i>	Chimpanzee	EN	155	135	8	2	91	0
	<i>Gorilla gorilla</i>	Western gorilla	CR	25	18	15	13	4	1
	<i>Colobus satanas</i>	Black colobus	VU	0	0	0	0	0	0
	<i>Colobus guereza</i>	Guereza colobus	LC	3	3	4	3	2	0
	<i>Mandrillus sphinx</i>	Mandrill	VU	104	2	4	1	2	0
	<i>Cercocebus torquatus</i>	Red-capped mangabey	VU	4	2	0	0	0	0
	<i>Cercocebus agilis</i>	Agile mangabey	LC	0	2	1	2	0	0
	<i>Lophocebus albigena</i>	Grey-cheeked mangabey	LC	1	10	1	1	0	1
	<i>Miopithecus ogouensis</i>	Gabon talapoin	LC	0	0	5	0	0	3
	<i>Cercopithecus solatus</i>	Sun-tailed monkey	VU	0	0	1	2	0	2
	<i>Cercopithecus neglectus</i>	De Brazza's monkey	LC	0	4	1	0	0	2
	<i>Cercopithecus pogonias</i>	Crowned monkey	LC	0	6	0	1	0	5
	<i>Cercopithecus nictitans</i>	Putty-nosed monkey	LC	1	7	1	1	0	4
	<i>Cercopithecus cephus</i>	Moustached monkey	LC	0	8	2	10	0	6
	<i>Perodicticus potto</i>	Potto	LC	3	1	2	1	3	0
	<i>Arctocebus aureus</i>	Golden angwantibo	LC	0	0	3	1	0	0
	<i>Euoticus elegantulus</i>	Western needle-clawed galago	LC	3	0	3	2	0	0
	<i>Galagoides demidoff</i>	Demidoff's galago	LC	3	0	8	2	0	0
	<i>Galago thomasi</i>	Thomas's galago	LC	0	0	0	0	0	0
	<i>Galago Alleni</i>	Allen's squirrel galago	LC	2	0	4	1	0	0
Carnivora	<i>Canis adustus</i>	Side-Striped jackal	LC	8	1	0	0	0	0
	<i>Mellivora capensis</i>	Honey badger	LC	0	0	0	0	0	0
	<i>Atilax paludinosus</i>	Marsh mongoose	LC	3	1	0	0	0	0
	<i>Bdeogale nigripes</i>	Blacked-legged mongoose	LC	1	0	0	0	0	0
	<i>Herpestes naso</i>	Long-snouted mongoose	LC	2	0	0	0	0	0
	<i>Herpestes sanguineus</i>	Slender mongoose	LC	14	0	0	0	0	0
	<i>Herpestes ichneumon</i>	Egyptian mongoose	LC	24	0	0	1	0	0
	<i>Crossarchus platycephalus</i>	Flat-headed cusimanse	LC	4	0	0	0	0	0
	<i>Crocuta crocuta</i>	Spotted hyena	LC	63	0	0	1	4	0
	<i>Civettictis civetta</i>	African civet	LC	7	1	0	0	0	0
	<i>Genetta tigrina</i>	Blotched genet	LC	14	0	0	0	0	0
	<i>Genetta servalina</i>	Servaline genet	LC	5	0	0	0	0	0
	<i>Nandinia binotata</i>	African palm civet	LC	8	1	0	0	0	0
	<i>Poiana richardsonii</i>	African linsang	LC	3	0	0	0	0	0
	<i>Leptailurus serval</i>	Serval	LC	0	0	0	1	0	0
	<i>Panthera pardus</i>	Leopard	NT	33	1	8	9	2	0
	<i>Caracal aurata</i>	African golden cat	NT	0	0	0	1	0	0
Pholidota	<i>Phataginus tricuspis</i>	Tree pangolin	NT	3	1	0	0	0	0
	<i>Uromanis tetradactyla</i>	Long-tailed pangolin	LC	0	0	0	0	2	0
	<i>Smutsia gigantea</i>	Giant pangolin	NT	0	0	0	0	0	0
Tubulidentata	<i>Orycteropus afer</i>	Aardvark	LC	1	2	2	2	2	0
Hyracoidea	<i>Dendrohyrax dorsalis</i>	Western tree hyrax	LC	0	0	1	0	2	0
Proboscidea	<i>Loxodonta africana</i>	African elephant	VU	120	0	0	0	3	0

Appendix 1. (Continued)

Order	Scientific name	Common name	IUCN Status	Cytb	COI	12s rRNA	16s rRNA	Mt genome	Partial Mt
Cetartiodactyla	<i>Hippopotamus amphibius</i>	Hippopotamus	VU	2	0	3	3	3	0
	<i>Hylochoerus meinertzhageni</i>	Giant forest hog	LC	1	0	1	1	0	0
	<i>Potamochoerus porcus</i>	Red river hog	LC	4	7	2	2	2	0
	<i>Hyemoschus aquaticus</i>	Water chevrotain	LC	2	0	0	0	2	0
	<i>Cephalophus callipygus</i>	Peters' duiker	LC	9	28	2	0	4	0
Cetartiodactyla	<i>Cephalophus dorsalis</i>	Bay duiker	LC	9	11	2	0	2	0
	<i>Cephalophus leucogaster</i>	White-bellied duiker	LC	4	11	1	0	2	0
	<i>Philantomba monticola</i>	Blue duiker	LC	50	32	3	0	3	0
	<i>Cephalophus nigrifrons</i>	Black-fronted duiker	LC	4	7	1	0	2	0
	<i>Cephalophus ogilbyi</i>	Ogilby's duiker	LC	4	1	1	0	2	0
	<i>Cephalophus silvicultor</i>	Yellow-backed duiker	LC	5	2	1	0	2	0
	<i>Kobus ellipsiprymnus</i>	Waterbuck	LC	7	4	2	2	2	0
	<i>Neotragus batesi</i>	Dwarf antelope	LC	0	0	0	0	2	0
	<i>Sylvicapra grimmia</i>	Bush duiker	LC	13	3	3	1	2	0
	<i>Syncerus caffer</i>	African buffalo	LC	34	13	2	2	45	0
	<i>Tragelaphus eurycerus</i>	Bongo	NT	4	2	2	3	2	0
	<i>Tragelaphus scriptus</i>	Bushbuck	LC	175	4	2	2	4	0
	<i>Tragelaphus spekii</i>	Sitatunga	LC	4	1	1	1	2	0